Consensus-Based Gene Selection for Molecular Subtype Classification and Exploratory Survival Analysis in the METABRIC Breast Cancer Cohort

October 10, 2026

Abstract

Molecular subtype characterization is an important task in breast cancer research, while survival modeling offers a complementary view of patient outcomes. This report documents a notebook-based analysis of 1,904 METABRIC-derived patient records and 490 candidate gene-related features. Ten complementary feature-importance or selection procedures—Random Forest importance, XGBoost importance, XGBoost gain, permutation importance, SHAP, LASSO, Elastic Net, mutual information, ANOVA F-statistics, and recursive feature elimination—are combined through min–max normalization, an arithmetic consensus score, and a top-50 recurrence count. The five candidate genes are *EGFR*, *CDK1*, *GATA3*, *ERBB2*, and *ERBB3*. The supplied notebook reports an XGBoost seven-class accuracy of 0.8346 and macro-F1 of 0.6780 using the full feature set. In a separate comparison, the best full-feature model among RBF-SVM, LightGBM, and Gaussian Naive Bayes is RBF-SVM (accuracy 0.7900; macro-F1 0.6756), whereas the best five-gene model is Gaussian Naive Bayes (accuracy 0.6955; macro-F1 0.5505). The five-gene panel therefore improves compactness and interpretability but does not preserve full-model predictive performance in this experiment. Exploratory Cox concordance increases from 0.6834 for clinical covariates to 0.6855 after adding the five genes ($\Delta C = 0.0021$); these are apparent in-sample estimates and not independent validation. A critical methodological concern is that test-set permutation and SHAP scores contribute to feature ranking, creating information leakage. The results should be interpreted as exploratory computational evidence. Nested validation, a genuinely untouched test set, a five-gene-only XGBoost experiment, rare-class-aware metrics, and external validation are required before any clinical interpretation.

Keywords: METABRIC, breast cancer, molecular subtype, gene selection, consensus ranking, XGBoost, SHAP, feature importance, Cox model, Kaplan–Meier.

Contents

1 Introduction **4**
- 1.1 Research questions . . . 4
- 1.2 Contributions . . . 4

2 Dataset and target definition **5**
- 2.1 Cohort overview . . . 5
- 2.2 Multiclass target . . . 5
- 2.3 Train/test split and preprocessing . . . 6

3 Methodology **7**
3.1 Problem formulation . . . 7
3.2 Feature Importance Analysis . . . 7
3.2.1 Random Forest feature importance . . . 8
3.2.2 XGBoost importance . . . 8
3.2.3 XGBoost gain . . . 8
3.2.4 Permutation importance . . . 9
3.2.5 SHAP importance . . . 9
3.2.6 LASSO coefficient importance . . . 9
3.2.7 Elastic Net coefficient importance . . . 10
3.2.8 Mutual information . . . 10
3.2.9 ANOVA F-statistic . . . 10
3.2.10 Recursive feature elimination . . . 10
3.3 Min–max normalization . . . 11
3.4 Consensus score and selection frequency . . . 11
3.5 Why use a consensus rather than one method? . . . 12

4 Survival analysis **12**
4.1 Event coding and analysis cohort . . . 12
4.2 Cox proportional-hazards model . . . 12
4.3 Concordance index . . . 13
4.4 Kaplan–Meier estimator . . . 13

5 Classifier and evaluation metrics **14**
5.1 XGBoost classifier . . . 14
5.2 Accuracy . . . 14
5.3 Precision, recall, and F1 . . . 14
5.4 Model comparison . . . 14

6 Results **15**
6.1 Consensus-selected genes and importance . . . 15
6.2 Cross-method evidence . . . 15
6.3 Exploratory survival analysis . . . 16
6.4 Subtype classification results . . . 17
6.5 Full 490-feature set versus five-gene panel . . . 20

7 Discussion **22**

8 Conclusion **22**

List of Figures

1 Subtype distribution in the analyzed cohort. The NC category contains only six observations, which makes reliable class-specific generalization estimates difficult. . . . 6
2 workflow of the study . . . 7
3 Reported consensus scores for the five selected genes. The plotted values are the saved consensus scores; final ranking additionally uses selection frequency. . . . 15
4 Cox concordance comparison. Both C-index values are calculated on the observations used to fit the corresponding model; they are apparent, in-sample estimates. . . . 16

5 Kaplan–Meier survival curves for high- versus low-expression groups of EGFR, split at the cohort median. . . . 17
6 Kaplan–Meier survival curves for high- versus low-expression groups of CDK1, split at the cohort median. . . . 17
7 Kaplan–Meier survival curves for high- versus low-expression groups of GATA3, split at the cohort median. . . . 18
8 Kaplan–Meier survival curves for high- versus low-expression groups of ERBB2, split at the cohort median. . . . 18
9 Kaplan–Meier survival curves for high- versus low-expression groups of ERBB3, split at the cohort median. . . . 19
10 Class-wise F1 scores from the reported XGBoost classification report. NC has only one test observation, so its zero score is not a stable estimate of performance for that class. . . . 20
11 Accuracy comparison for the three evaluated models using all 490 features and the selected five-gene panel. . . . 21
12 Macro-F1 comparison for the full feature set and five-gene panel. . . . 21

List of Tables

1 Dataset overview as reported by the notebook. . . . 5
2 Observed subtype class distribution. . . . 6
3 Five candidate genes and their reported consensus scores. . . . 15
4 Apparent in-sample Cox concordance values. . . . 16
5 Reported XGBoost results on the held-out test split. . . . 19
6 Reported classification comparison: full feature set versus five-gene panel. . . . 20

1 Introduction

Breast cancer comprises biologically heterogeneous disease groups. Molecular subtype labels such as Basal, HER2-enriched (Her2), Luminal A (LumA), Luminal B (LumB), Normal-like (Normal), and claudin-low represent different expression patterns and may be associated with distinct clinical trajectories. The dataset used here contains a seven-label target, including a rare label named NC. Predicting these labels from a broad genomic feature matrix is a multiclass learning problem, not a binary cancer/non-cancer task.

High-dimensional data contain correlated and potentially redundant variables. A single feature-importance measure can be biased toward a model family, a feature scale, or a particular definition of importance. Tree split importance, gain, permutation importance, SHAP attribution, sparse linear coefficients, univariate association, and recursive elimination answer related but not identical questions. A consensus framework can identify candidates that recur across assumptions. However, consensus is not automatically a guarantee of biological causality or clinical validity.

The analysis has two distinct objectives:

1. **Subtype classification and feature selection:** rank 490 candidate gene-related features and derive a compact five-gene panel; evaluate multiclass prediction using held-out records.
2. **Exploratory survival analysis:** compare Cox proportional-hazards discrimination using clinical covariates alone versus clinical covariates augmented with the five selected genes.

The two objectives must not be conflated. A feature useful for subtype classification need not add survival discrimination after clinical covariates are considered. Likewise, a gene with high predictive importance is not necessarily a statistically significant independent prognostic factor.

1.1 Research questions

The study addresses the following questions:

- Which candidate genes are repeatedly ranked highly by diverse selection measures?
- How does a compact five-gene feature set compare with the full 490-feature set for subtype prediction?
- Are reported errors concentrated in underrepresented subtype classes?
- Does adding the selected genes improve the apparent concordance of a clinical Cox model?
- What methodological changes are needed for an unbiased and reproducible evaluation?

1.2 Contributions

The major contributions of this work are summarized as follows:

1. **Consensus-Based Feature Selection:** Develop a feature selection framework combining multiple selection techniques to identify informative genomic features from the high-dimensional METABRIC dataset.
2. **Feature Importance Analysis:** Investigate five selected candidate genes, namely EGFR, CDK1, GATA3, ERBB2, and ERBB3, to assess their contributions to breast cancer molecular subtype classification.

3. **Multiclass Subtype Classification:** Evaluate machine learning models for molecular subtype prediction using accuracy, precision, recall, and macro-F1 score.

4. **Survival Prognosis Analysis:** Explore the association between gene expression and patient survival using Kaplan–Meier curves, Cox proportional hazards modelling, and the concordance index.

5. **Interpretable Genomic Analysis:** Integrate feature importance, subtype classification, and survival analysis to identify promising genomic markers for further investigation in breast cancer prognosis.

2 Dataset and target definition

2.1 Cohort overview

The notebook reads `METABRIC_RNA_Mutation.csv`. The dataset contains 1,904 rows and 693 columns before feature selection. The workflow excludes clinical covariates, mutation features ending in `_mut`, identifiers, survival outcomes, and target columns from the candidate matrix. This leaves 490 candidate gene-related variables. The manuscript's use of the term "gene features" follows the notebook; the set is not necessarily a pure transcriptomic matrix because it includes variables such as `mutation_count` and named gene-level columns.

Table 1: Dataset overview as reported by the notebook.

Quantity	Value
Patient records	1,904
Original columns	693
Candidate features used for ranking	490
Clinical variables considered in survival analysis	21
Training records (80%)	1,523
Held-out test records (20%)	381
Observed subtype classes	7
Records retained for survival analysis	1,903

2.2 Multiclass target

The target is `pam50_+_claudin-low_subtype`. Its observed labels and counts are shown in Table 2. Although a five-class task may have been contemplated in an earlier framing, the executed notebook has seven classes. Results in this report therefore describe seven-class classification.

Table 2: Observed subtype class distribution.

Subtype	Patients	Percentage
LumA	679	35.66%
LumB	461	24.21%
Her2	220	11.55%
claudin-low	199	10.45%
Basal	199	10.45%
Normal	140	7.35%
NC	6	0.32%
Total	1,904	100%

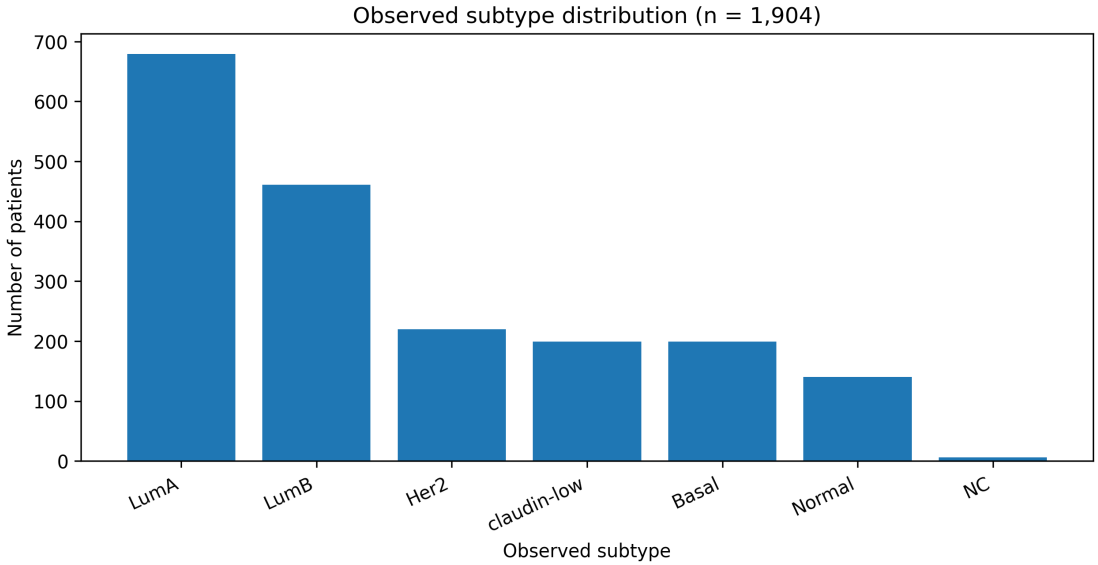


Figure 1: Subtype distribution in the analyzed cohort. The NC category contains only six observations, which makes reliable class-specific generalization estimates difficult.

The largest class, LumA, has 679 records, whereas NC has only six. Stratification preserves approximate class proportions in the train/test split, but it cannot create information that is absent. The test set contains one NC example, as reflected in the classification report. Consequently, class-specific estimates for NC are unstable and should not be interpreted as a robust estimate of performance. Overall accuracy is dominated by common classes; macro-averaged metrics are more sensitive to rare-class failure but themselves become unstable when a class has only one test example.

2.3 Train/test split and preprocessing

In this study we apply a stratified 80:20 split with random state 42. The training matrix has dimensions 1523×490 and the test matrix 381×490. Median imputation is fitted on the training matrix and applied to both partitions. Standard scaling is fitted on the imputed training matrix for methods requiring standardized values, including coefficient-based methods and RFE. Tree-based models are trained on the imputed, unscaled matrix.

For feature j, the training median is

$$m_j = \text{median}\{x_{ij} : i \in \mathcal{D}_{\text{train}},\ x_{ij} \text{ observed}\}.$$

The imputed value is

$$\tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \text{ observed}, \\ m_j, & \text{otherwise}. \end{cases}$$

Standardization is

$$z_{ij} = \frac{\tilde{x}_{ij} - \mu_j}{\sigma_j},$$

where μ_j and σ_j are estimated from the training partition only.

3 Methodology

3.1 Problem formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, with $N = 1904$, $\mathbf{x}_i \in \mathbb{R}^p$, $p = 490$, and $y_i \in \{1, \ldots, K\}$, $K = 7$. The classification task estimates

$$\hat{f} : \mathbb{R}^{490} \to \{1, \ldots, 7\}.$$

The feature-selection task estimates a ranked list $\pi = (\pi_1, \ldots, \pi_p)$ and selects a panel $\mathcal{G}_5 = \{\pi_1, \ldots, \pi_5\}$. The survival task uses duration T_i and event indicator δ_i with clinical covariates $\mathbf{c}_i$ and selected genes $\mathbf{g}_i$.

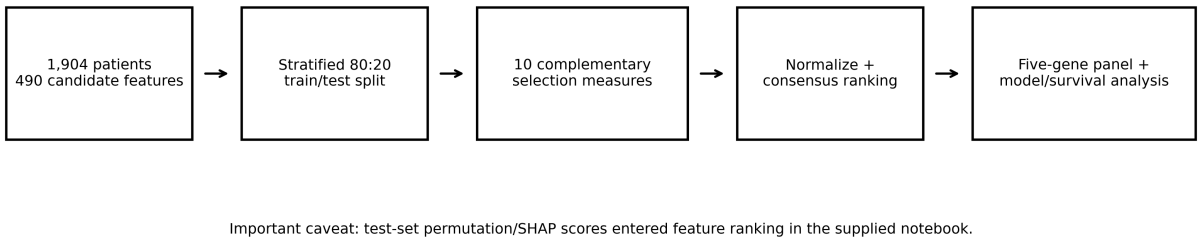


Figure 2: workflow of the study

3.2 Feature Importance Analysis

To identify the most informative genomic features for breast cancer molecular subtype classification, multiple feature selection and importance estimation methods were employed. These methods capture different aspects of feature relevance, including model-based importance, statistical association, coefficient-based selection, and prediction-level contribution. The investigated techniques include Random Forest feature importance, XGBoost gain-based importance, permutation importance, SHAP (SHapley Additive exPlanations), LASSO, Elastic Net, mutual information, ANOVA F-statistic, and Recursive Feature Elimination (RFE). The importance scores obtained from the individual methods were analyzed to compare the consistency of feature rankings and identify potentially informative genes. To facilitate comparison across methods with different score ranges, the importance scores were normalized before computing a consensus ranking. Based on the resulting feature-selection analysis, five candidate genes—EGFR, CDK1, GATA3, ERBB2, and ERBB3—were selected for further investigation. Their importance was examined in relation to molecular subtype classification and exploratory survival analysis. The use of multiple feature selection methods reduces dependence on a single algorithm and provides complementary perspectives on genomic

feature relevance. However, feature importance is model- and dataset-dependent and does not establish a causal relationship or clinical significance. Therefore, the selected genes are treated as candidate genomic markers requiring further validation. For an unbiased assessment of predictive performance, feature selection and importance estimation should be performed exclusively within the training data during cross-validation, with final evaluation on an independent test set.

3.2.1 Random Forest feature importance

A Random Forest consists of B decision trees. At each split, a feature is selected to reduce node impurity. For a node t split by feature j, the weighted impurity decrease can be written as

$$\Delta I(t, j) = \frac{n_t}{N}\left[I(t) - \frac{n_{t_L}}{n_t} I(t_L) - \frac{n_{t_R}}{n_t} I(t_R)\right],$$

where $I(\cdot)$ is an impurity measure and n_t, n_{t_L}, n_{t_R} are node sample counts. The forest importance is the normalized sum of impurity reductions attributed to feature j:

$$FI_j^{RF} = \frac{\sum_{b=1}^{B} \sum_{t \in \mathcal{T}_b:\, v(t)=j} \Delta I(t, j)}{\sum_{\ell=1}^{p} \sum_{b=1}^{B} \sum_{t \in \mathcal{T}_b:\, v(t)=\ell} \Delta I(t, \ell)}.$$

The notebook uses 500 trees, random state 42, and `class_weight=balanced`. This method measures how much a feature helps reduce impurity in the fitted forest. It may favor variables with many potential split points and can distribute importance across correlated predictors.

3.2.2 XGBoost importance

XGBoost builds an additive ensemble of trees:

$$\hat{y}_i = \sum_{k=1}^{K_T} f_k(\mathbf{x}_i), \qquad f_k \in \mathcal{F},$$

where K_T is the number of boosting rounds and $\mathcal{F}$ is the space of regression trees. Its regularized objective is commonly written as

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \Omega(f) = \gamma T + \frac{\lambda}{2} \sum_{j=1}^{T} w_j^2 + \alpha \sum_{j=1}^{T} |w_j|.$$

Here T is the number of leaves, w_j are leaf scores, and γ, λ, α control model complexity. For multiclass classification, the notebook uses `multi:softprob`, 300 estimators, maximum depth 5, learning rate 0.05, row subsampling 0.8, column subsampling 0.8, and multiclass log loss. The default feature importance returned by the fitted XGBoost model is distinct from its explicit gain score.

3.2.3 XGBoost gain

Gain measures the loss reduction attributable to splits using a feature. For a split with gradient sums G_L, G_R and Hessian sums H_L, H_R, a regularized gain expression is

$$\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\right] - \gamma.$$

The notebook queries `get_score(importance_type='gain')`. Features not used in a split are assigned zero by the dictionary lookup. The score is therefore sparse and measures split utility within

this particular boosted-tree fit, not an intrinsic biological effect.

3.2.4 Permutation importance

Permutation importance measures the deterioration in a chosen predictive score after randomly permuting one feature. Let $S(f, X, y)$ be the macro-F1 score and let $X^{\pi(j)}$ denote the data with column j permuted. Then

$$PI_j = S(f, X, y) - \frac{1}{R}\sum_{r=1}^{R} S(f, X^{\pi_r(j)}, y).$$

Positive values indicate that permutation generally reduces model performance; values near zero indicate limited unique contribution under this model and scoring rule. Negative values can occur due to sampling noise or when disrupting a variable slightly improves the score. The notebook uses macro-F1, ten repeats, and random state 42. Critically, it applies this calculation to the held-out test set, so this information should not be used to select final features if that test set is later used to report performance.

3.2.5 SHAP importance

SHAP assigns each feature a contribution to a particular prediction based on Shapley values. For feature j and observation x, the Shapley value is

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} \left[f_x(S \cup \{j\}) - f_x(S) \right].$$

Here F is the complete feature set and $f_x(S)$ denotes the model output when features outside S are integrated out according to the SHAP background formulation. The global mean absolute SHAP importance is

$$I_j^{SHAP} = \frac{1}{n}\sum_{i=1}^{n} |\phi_j(\mathbf{x}_i)|.$$

For multiclass output, the notebook aggregates absolute values across observations and classes. SHAP is a model-attribution method: it explains the fitted model's predictions, not causal influence. The notebook's SHAP scores are based on test observations, which contributes to the leakage concern.

3.2.6 LASSO coefficient importance

LASSO uses an ℓ_1 penalty to encourage sparse coefficients. For a multiclass logistic model, the objective can be expressed as

$$\min_{\mathbf{B}} \; -\sum_{i=1}^{n} \log P(y_i \mid \mathbf{x}_i; \mathbf{B}) + \lambda \sum_{k=1}^{K}\sum_{j=1}^{p} |\beta_{kj}|.$$

The notebook uses logistic regression with the `saga` solver, $C = 0.1$, and up to 3,000 iterations. It converts coefficients to a global feature score by averaging absolute coefficients over classes:

$$I_j^{LASSO} = \frac{1}{K}\sum_{k=1}^{K} |\beta_{kj}|.$$

Since coefficient magnitudes depend on scale, the features are standardized for this procedure. LASSO can shrink coefficients to zero, but correlated genes may substitute for one another across samples.

3.2.7 Elastic Net coefficient importance

Elastic Net combines ℓ_1 and ℓ_2 penalties:

$$\min_{\mathbf{B}} \; -\sum_{i=1}^{n} \log P(y_i \mid \mathbf{x}_i; \mathbf{B}) + \lambda \left[\rho \|\mathbf{B}\|_1 + \frac{1-\rho}{2} \|\mathbf{B}\|_2^2 \right].$$

The notebook uses $C = 0.1$, `l1_ratio`=0.5, and the `saga` solver. Its feature score is

$$I_j^{EN} = \frac{1}{K} \sum_{k=1}^{K} |\beta_{kj}|.$$

The ℓ_2 component can stabilize groups of correlated features, while the ℓ_1 component promotes sparsity. Like LASSO, this score is predictive and depends on the coding, scaling, and regularization strength.

3.2.8 Mutual information

Mutual information quantifies statistical dependence between feature X_j and class label Y:

$$I(X_j; Y) = \sum_{x_j} \sum_{y} p(x_j, y) \log \frac{p(x_j, y)}{p(x_j)p(y)}.$$

For continuous variables, the estimator approximates the dependence using local neighborhoods. Mutual information can capture nonlinear associations that a linear correlation measure misses. It is univariate: it does not account for conditional redundancy among genes or interactions with other features. The notebook uses `mutual_info_classif` with random state 42.

3.2.9 ANOVA F-statistic

The one-way ANOVA statistic compares between-class variation to within-class variation. For feature j and K classes:

$$F_j = \frac{\sum_{k=1}^{K} n_k (\bar{x}_{kj} - \bar{x}_j)^2 / (K-1)}{\sum_{k=1}^{K} \sum_{i:y_i=k} (x_{ij} - \bar{x}_{kj})^2 / (N-K)}.$$

Here $\bar{x}_{kj}$ is the class-specific mean, $\bar{x}_j$ the overall mean, and n_k the class size. Large F_j values indicate that class means differ relative to within-class variation. This is a univariate screening statistic and does not establish causal relevance. The notebook replaces non-finite values with zero before sorting.

3.2.10 Recursive feature elimination

RFE repeatedly fits an estimator, ranks features by estimator-derived importance, and removes a subset of the least important variables until the target count is reached. In this notebook, RFE uses logistic regression with an ℓ_2 penalty and selects 20 features from the 100 leading XGBoost

candidates. The indicator score is

$$I_j^{RFE} = \begin{cases} 1, & j \text{ retained by RFE,} \\ 0, & \text{otherwise.} \end{cases}$$

RFE therefore contributes a binary selection signal rather than a continuous effect size. Because it is applied to a prefiltered top-100 set, it is not an exhaustive search over all possible subsets of 490 variables.

3.3 Min–max normalization

The ten methods use different scales and units. For method m, let s_{jm} be the raw score for feature j. The normalized score is

$$\tilde{s}_{jm} = \begin{cases} \dfrac{s_{jm} - \min_\ell s_{\ell m}}{\max_\ell s_{\ell m} - \min_\ell s_{\ell m}}, & \max_\ell s_{\ell m} > \min_\ell s_{\ell m}, \\ 0, & \text{otherwise.} \end{cases}$$

This maps nonconstant method scores to $[0, 1]$. It makes the score ranges comparable, but does not make the methods statistically equivalent. A single extreme score can compress the normalized values of other features. For methods with negative permutation importance, the minimum is still mapped to zero and maximum to one.

3.4 Consensus score and selection frequency

For $M = 10$ methods, the arithmetic consensus score is

$$C_j = \frac{1}{M} \sum_{m=1}^{M} \tilde{s}_{jm}.$$

The notebook also counts how often each feature appears among the top 50 features for each method:

$$F_j = \sum_{m=1}^{M} \mathbb{I}\{j \in \text{Top50}(s_{\cdot m})\}, \qquad 0 \le F_j \le 10.$$

The final ranking sorts by F_j in descending order and uses C_j as a tie-breaker. This is a two-part ranking: recurrence rewards cross-method stability, while the consensus score distinguishes candidates with similar recurrence. Selection frequency and consensus score should be reported separately because they convey different information.

Algorithm 1 Consensus-based candidate gene selection

Require: Training data X_{train}, labels y_{train}, feature set G of size 490, methods $m = 1, \ldots, 10$

for each method m **do**
    Fit or evaluate the method according to its own objective
    Obtain raw scores s_{jm} for all candidate features
    Normalize the scores to $\tilde{s}_{jm} \in [0, 1]$
    Record the method-specific top-50 set
end for
for each feature j **do**
    $C_j \leftarrow \frac{1}{10} \sum_{m=1}^{10} \tilde{s}_{jm}$
    $F_j \leftarrow \sum_{m=1}^{10} \mathbb{I}\{j \in \text{Top50}_m\}$
end for
Sort features by descending F_j, then descending C_j
Return the first five features as candidate panel

3.5 Why use a consensus rather than one method?

Each method defines relevance differently. Random Forest and XGBoost assess contribution to tree-based prediction; gain measures split-level loss reduction; permutation importance tests reliance of a fitted model on a feature; SHAP decomposes model outputs; LASSO and Elastic Net estimate regularized linear predictive coefficients; mutual information and ANOVA assess marginal feature-label association; RFE evaluates iterative removal under an estimator. Agreement across these views can reduce dependence on a single method's inductive bias. Disagreement is also informative: it can reveal correlated substitutes, nonlinear effects, or model-specific preferences.

The consensus score is not a probability that a gene is biologically important. It is a normalized aggregate of ranking statistics. The use of min–max normalization and equal weights is a design choice, not a uniquely optimal formula. Sensitivity analyses should test alternative aggregation rules, such as rank aggregation, robust medians, or weights estimated entirely within training folds.

4 Survival analysis

4.1 Event coding and analysis cohort

Survival time is `overall_survival_months`. The notebook creates an event indicator as

$$\delta_i = 1 - \texttt{overall_survival}_i,$$

so $\delta_i = 1$ denotes the event according to the notebook's coding, and $\delta_i = 0$ denotes right censoring. Removing records with missing or nonpositive duration leaves 1,903 observations. The notebook reports 1,103 events and 801 censored records at the original cohort level; the survival fit reports 800 right-censored observations after duration filtering. This small count difference reflects the filtering step and should be checked when reproducing the notebook.

4.2 Cox proportional-hazards model

The Cox model expresses the hazard at time t for covariate vector $\mathbf{x}_i$ as

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}_i),$$

where $h_0(t)$ is the baseline hazard and $\boldsymbol{\beta}$ is a vector of coefficients. The hazard ratio for a one-unit increase in covariate j is

$$HR_j = \exp(\beta_j).$$

A hazard ratio above one indicates a higher modeled hazard conditional on the other covariates; below one indicates a lower modeled hazard. Interpretation depends on feature coding and units. The notebook uses an L2 penalizer of 0.1 to mitigate instability from collinearity. The partial log-likelihood, ignoring ties for compact notation, is

$$\ell(\boldsymbol{\beta}) = \sum_{i:\delta_i=1} \left[\boldsymbol{\beta}^\top \mathbf{x}_i - \log \left(\sum_{j:T_j \geq T_i} \exp(\boldsymbol{\beta}^\top \mathbf{x}_j) \right) \right].$$

The penalized estimate approximately maximizes

$$\ell_{\text{pen}}(\boldsymbol{\beta}) = \ell(\boldsymbol{\beta}) - \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2, \qquad \lambda = 0.1.$$

The baseline model uses 21 clinical covariates after one-hot encoding categorical variables. The augmented model adds the five selected genes.

4.3 Concordance index

The concordance index measures whether predicted risk correctly orders pairs of patients with comparable observed survival times. In simplified notation,

$$C = \frac{\sum_{(i,j)\in\mathcal{P}} \mathbb{I}(\hat{r}_i > \hat{r}_j)}{|\mathcal{P}|},$$

where $\mathcal{P}$ is the set of comparable patient pairs and $\hat{r}_i$ is a risk score, with ties typically handled by a defined convention. A value near 0.5 indicates discrimination close to chance; higher values indicate better ranking. C-index does not assess calibration and should be accompanied by uncertainty intervals and external or cross-validated estimates.

4.4 Kaplan–Meier estimator

For descriptive survival curves, each selected gene is split into high and low groups at its cohort median. The Kaplan–Meier estimator is

$$\widehat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j} \right),$$

where t_j are event times, d_j is the number of events at t_j, and n_j is the number at risk immediately before t_j. The notebook produces median-split curves for the five genes. Visual separation alone does not establish statistical significance; the supplied workflow does not report log-rank tests or multiplicity-adjusted survival comparisons. Median dichotomization also discards continuous expression information and may obscure nonlinear associations.

5 Classifier and evaluation metrics

5.1 XGBoost classifier

The notebook's primary XGBoost classifier uses 300 boosting estimators, maximum depth 5, learning rate 0.05, subsample 0.8, column subsampling 0.8, objective `multi:softprob`, and evaluation metric `mlogloss`. The predicted class is

$$\hat{y}_i = \arg\max_{k \in \{1,\ldots,K\}} \hat{p}_{ik},$$

where $\hat{p}_{ik}$ is the predicted probability for class k. Multiclass log loss is

$$\mathcal{L}_{\text{multi}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{I}(y_i = k) \log \hat{p}_{ik}.$$

5.2 Accuracy

Accuracy is the proportion of correctly classified observations:

$$\text{Accuracy} = \frac{\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i)}{N}.$$

It is intuitive but can be dominated by prevalent classes in imbalanced datasets.

5.3 Precision, recall, and F1

For a given class treated as positive versus all other classes:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN},$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Macro-F1 is the unweighted mean of the class-wise F1 scores:

$$F_{1,\text{macro}} = \frac{1}{K} \sum_{k=1}^{K} F_{1,k}.$$

Weighted-F1 weights each class score by its support:

$$F_{1,\text{weighted}} = \sum_{k=1}^{K} \frac{n_k}{N} F_{1,k}.$$

Macro precision and macro recall are defined analogously by averaging per-class precision and recall. In this task, macro-F1 is particularly important because it prevents LumA's large sample count from completely dominating the summary. However, an NC test support of one means the class-specific F1 estimate is extremely unstable.

5.4 Model comparison

The notebook includes a separate comparison of RBF-SVM, LightGBM, and Gaussian Naive Bayes using either the full 490-feature set or the selected five genes. This comparison should be distinguished from the earlier XGBoost result. The XGBoost score of 0.8346 accuracy and 0.6780 macro-F1 was obtained using the full feature set, not a five-gene-only model.

6 Results

6.1 Consensus-selected genes and importance

The final candidate panel reported by the notebook is *EGFR*, *CDK1*, *GATA3*, *ERBB2*, and *ERBB3*. Table 3 gives the consensus scores available in the saved analysis. The final ordering is based first on top-50 selection frequency and then on consensus score; accordingly, the rank order is not simply the descending order of the displayed consensus values.

Table 3: Five candidate genes and their reported consensus scores.

Final rank	Gene	Consensus score	Interpretation
1	EGFR	0.809863	Highest reported consensus score; selected by all ten methods
2	CDK1	0.567962	Selected by all ten methods; strong cross-method recurrence
3	GATA3	0.665170	High consensus score; selected by nine methods
4	ERBB2	0.502867	Member of the final five-gene panel
5	ERBB3	0.449620	Member of the final five-gene panel

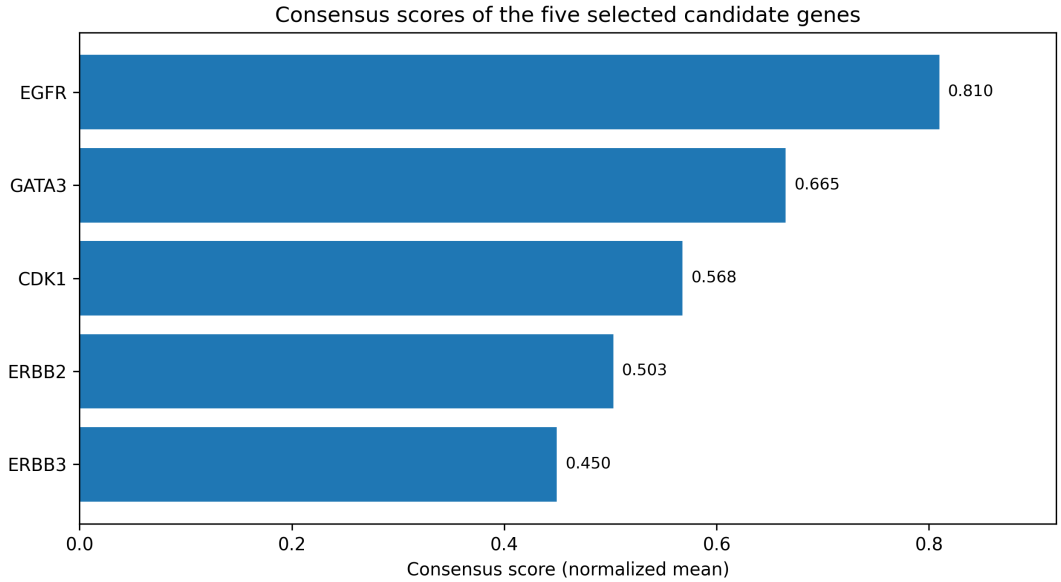


Figure 3: Reported consensus scores for the five selected genes. The plotted values are the saved consensus scores; final ranking additionally uses selection frequency.

Important ranking nuance. EGFR has the largest reported consensus score, while CDK1 is placed second because the final sort prioritizes selection frequency before consensus score. GATA3's score (0.665170) exceeds CDK1's (0.567962), but GATA3 is ranked third because its top-50 recurrence is nine methods rather than ten. This distinction is essential when interpreting the selection algorithm. The full feature-importance CSV should be used to report verified frequencies for ERBB2 and ERBB3; these are not inferred here when the exact counts are not unambiguously visible in the extracted outputs.

6.2 Cross-method evidence

The outputs show that EGFR, CDK1, GATA3, ERBB2, and ERBB3 recur in several methods, but competing genes such as CSF1R, AURKA, CCNE1, MAPT, CHEK1, and LAMB3 also rank

highly under individual criteria. Examples from the notebook include EGFR leading the Random Forest importance table, GATA3 leading the XGBoost importance table, EGFR leading permutation importance, EGFR leading SHAP importance, EGFR/CDK1/ERBB2 appearing near the top of LASSO and Elastic Net, and GATA3/EGFR leading mutual information. This agreement supports prioritizing the five genes for further study, but disagreement is expected because the measures quantify different properties.

6.3 Exploratory survival analysis

The clinical-only Cox model reports $C = 0.6834217$. Adding the five selected genes yields $C = 0.6855178$. The difference is

$$\Delta C = 0.6855178 - 0.6834217 = 0.0020961 \approx 0.0021.$$

Table 4: Apparent in-sample Cox concordance values.

Model	C-index
Clinical covariates only	0.6834
Clinical covariates + five genes	0.6855
Absolute difference	0.0021

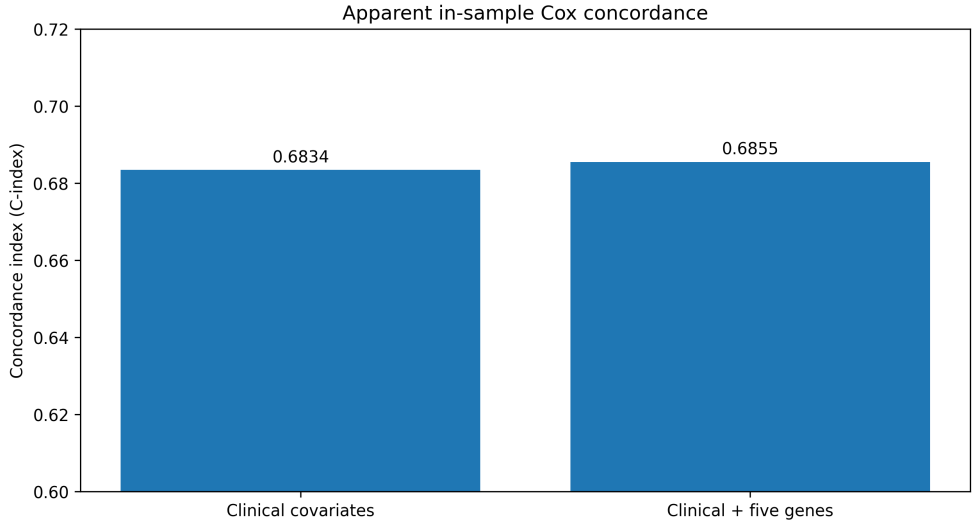


Figure 4: Cox concordance comparison. Both C-index values are calculated on the observations used to fit the corresponding model; they are apparent, in-sample estimates.

The increase is numerically small and cannot be called statistically significant without uncertainty estimates and a valid comparison procedure. The notebook reports the following gene coefficient p-values in the augmented Cox model: EGFR, 0.565; CDK1, 0.152; GATA3, 0.454; ERBB2, 0.440; and ERBB3, 0.393. None is below the conventional 0.05 threshold. These results do not establish independent prognostic associations. The models are fitted and evaluated on the same observations, so their concordance may be optimistic. The top-five consensus importance chart, five median-split Kaplan–Meier curves, and eight method-specific feature-importance charts. They are included here in the Results and Discussion section as requested. The Kaplan–Meier figures are descriptive; no inferential conclusion should be drawn from visual separation alone. The method-specific importance

plots are useful for examining agreement and disagreement between selection criteria, but their raw importance scales differ and should not be compared directly across methods.

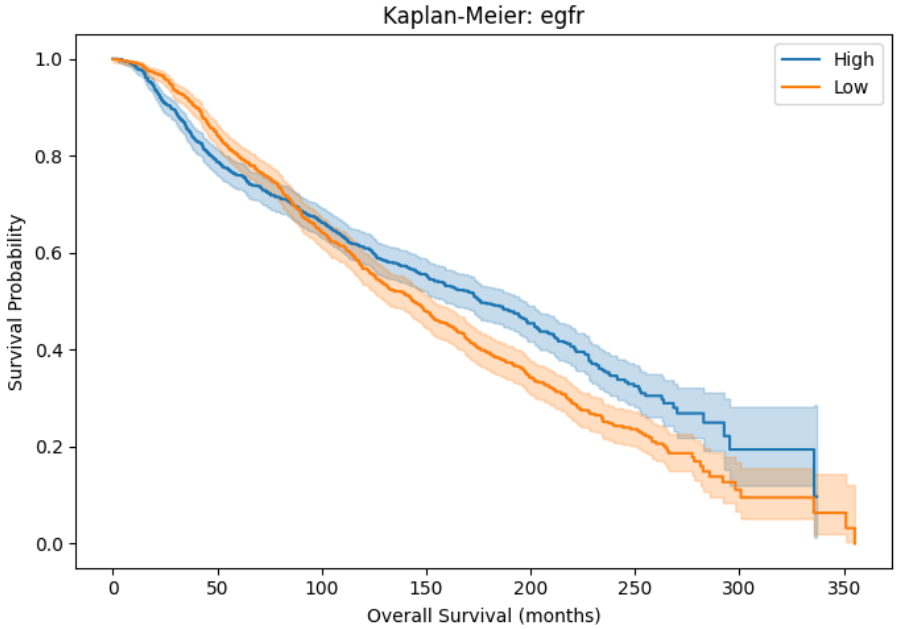


Figure 5: Kaplan–Meier survival curves for high- versus low-expression groups of EGFR, split at the cohort median.

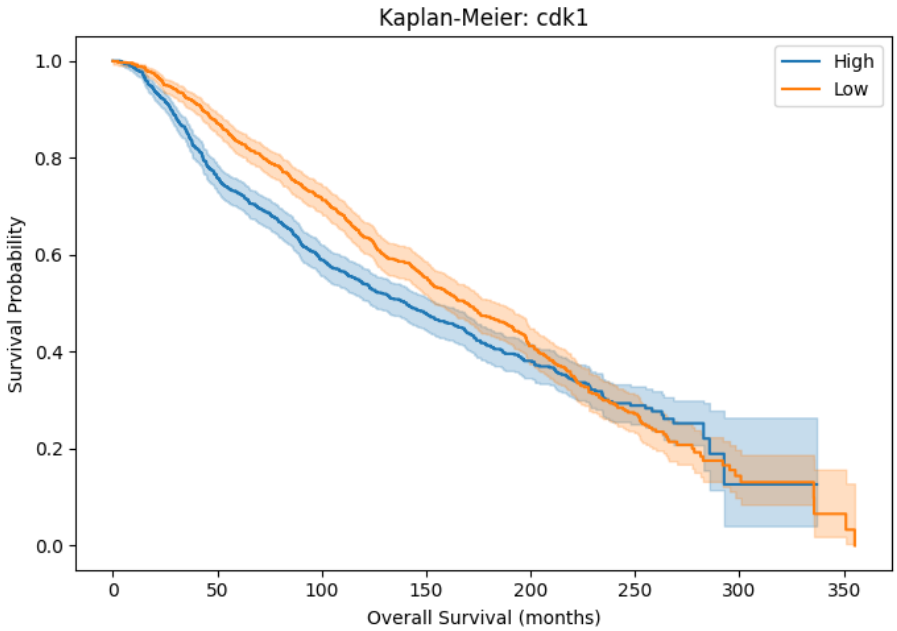


Figure 6: Kaplan–Meier survival curves for high- versus low-expression groups of CDK1, split at the cohort median.

6.4 Subtype classification results

The primary XGBoost experiment used all 490 candidate features. The notebook reports test accuracy 0.8346 and macro-F1 0.6780 on 381 records. The class-level report is summarized in Table 5. The difference between accuracy and macro-F1 is consistent with unequal class sizes and

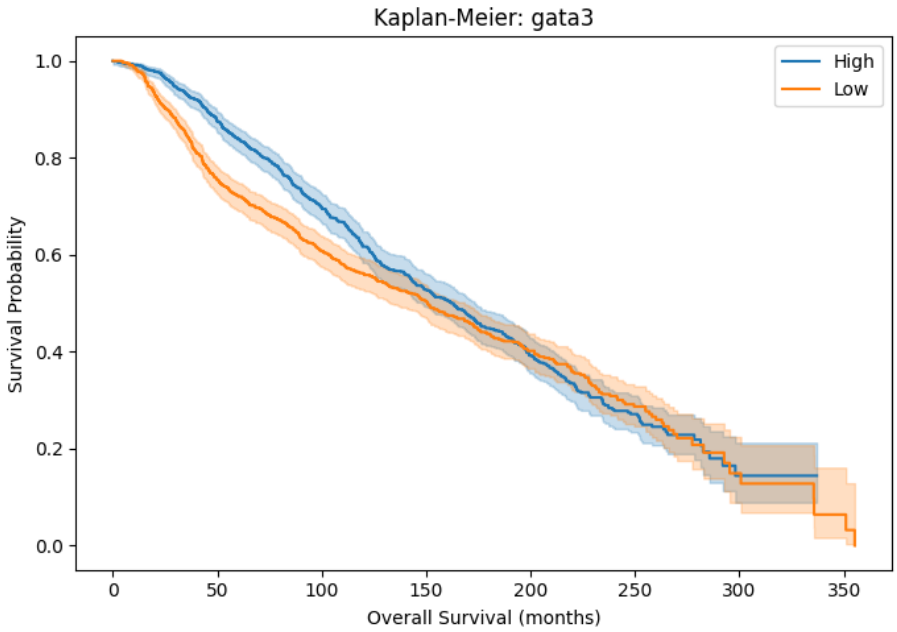


Figure 7: Kaplan–Meier survival curves for high- versus low-expression groups of GATA3, split at the cohort median.

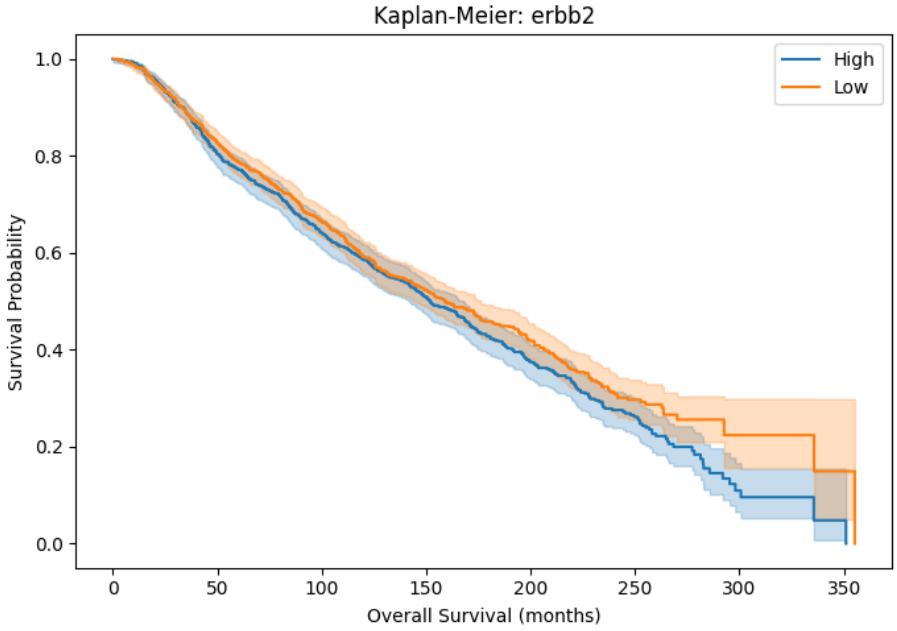


Figure 8: Kaplan–Meier survival curves for high- versus low-expression groups of ERBB2, split at the cohort median.

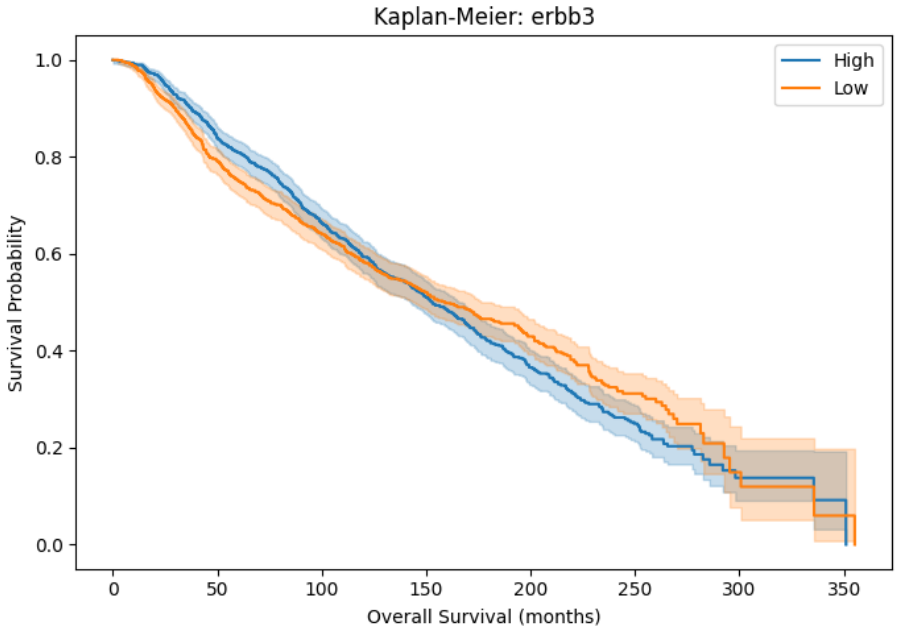


Figure 9: Kaplan–Meier survival curves for high- versus low-expression groups of ERBB3, split at the cohort median.

weaker results for NC and Normal.

Table 5: Reported XGBoost results on the held-out test split.

Class	Precision	Recall	F1	Support
Basal	0.88	0.90	0.89	40
Her2	0.84	0.84	0.84	44
LumA	0.83	0.90	0.87	136
LumB	0.84	0.86	0.85	92
NC	0.00	0.00	0.00	1
Normal	0.67	0.36	0.47	28
claudin-low	0.85	0.82	0.84	40
Accuracy		0.8346		381
Macro-F1		0.6780		381

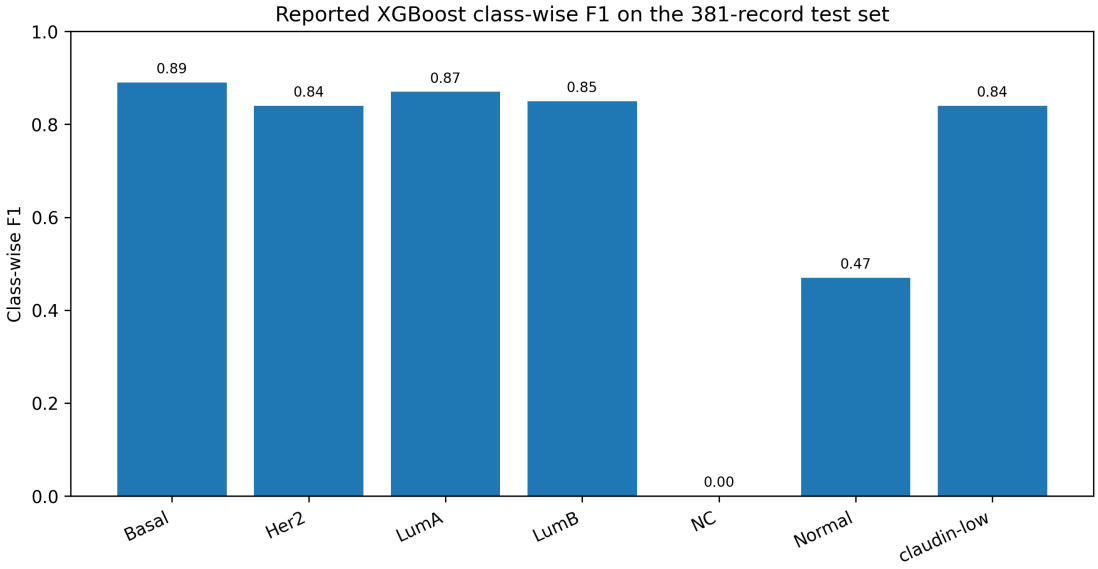


Figure 10: Class-wise F1 scores from the reported XGBoost classification report. NC has only one test observation, so its zero score is not a stable estimate of performance for that class.

6.5 Full 490-feature set versus five-gene panel

A separate experiment evaluates RBF-SVM, LightGBM, and Gaussian Naive Bayes on the full feature set and the five selected genes. Table 6 records the output values. In the full-feature setting, LightGBM has the highest accuracy (0.8189) among the three listed models, while RBF-SVM has the highest macro-F1 (0.6756). The notebook's summary selects RBF-SVM as the best full-feature model based on its comparison criteria. For the five-gene panel, Gaussian Naive Bayes has the highest accuracy (0.6955) and macro-F1 (0.5505) among the three models.

Table 6: Reported classification comparison: full feature set versus five-gene panel.

Feature set	Model	Accuracy	Macro precision	Macro recall	Macro-F1
Full	RBF-SVM	0.7900	0.6717	0.6845	0.6756
Full	LightGBM	0.8189	0.6855	0.6706	0.6755
Full	Gaussian NB	0.7113	0.5789	0.5969	0.5816
Five genes	RBF-SVM	0.6247	0.5577	0.5500	0.5429
Five genes	LightGBM	0.6798	0.5466	0.5550	0.5496
Five genes	Gaussian NB	0.6955	0.5802	0.5372	0.5505

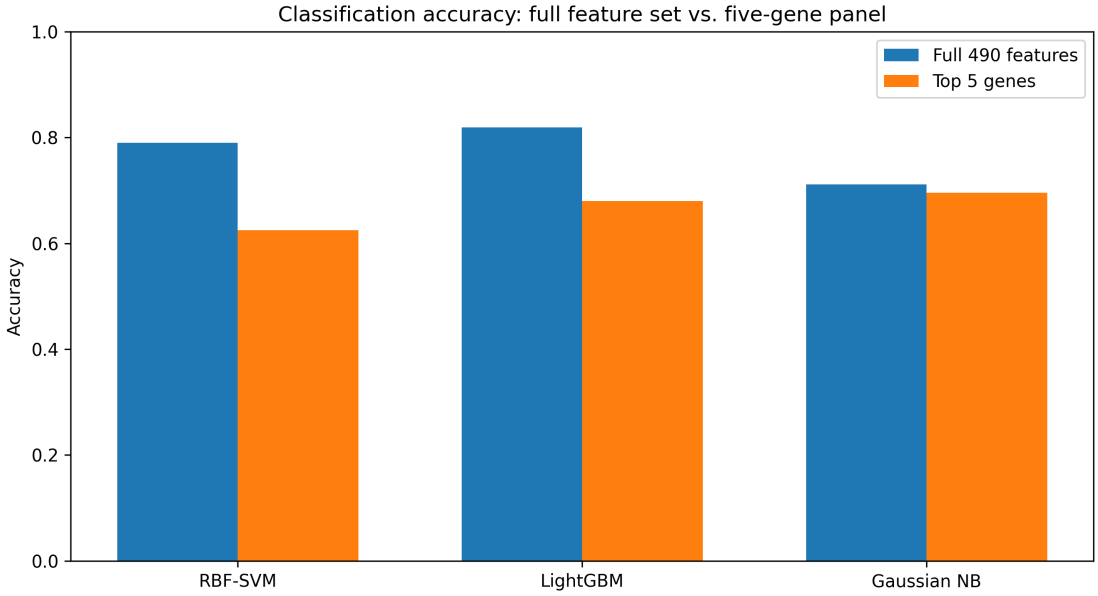


Figure 11: Accuracy comparison for the three evaluated models using all 490 features and the selected five-gene panel.

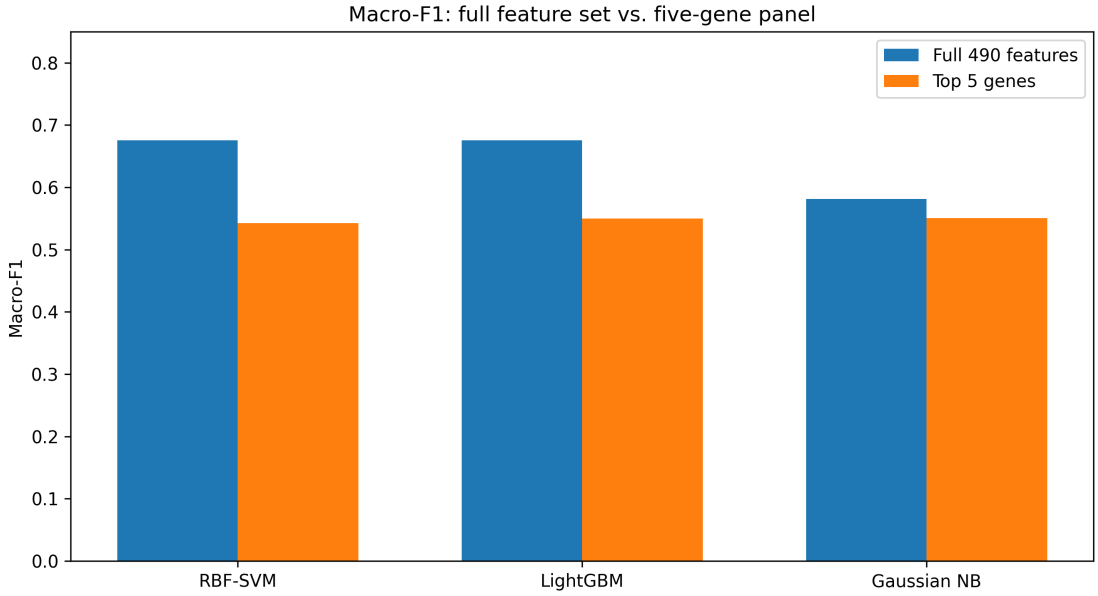


Figure 12: Macro-F1 comparison for the full feature set and five-gene panel.

The five-gene panel reduces the number of input features from 490 to five, a 98.98% reduction in dimensionality:

$$\text{Reduction} = \frac{490 - 5}{490} \times 100\% = 98.98\%.$$

This improves compactness and may reduce acquisition or computation burden in a future targeted assay, although actual clinical assay cost was not evaluated. In the supplied model comparison, the best five-gene macro-F1 (0.5505) is below the best full-feature macro-F1 (0.6756), a difference of approximately 0.1251. Thus, the current result supports a trade-off: the panel is more compact and interpretable, but it loses multiclass predictive information relative to the broader feature set.

7 Discussion

The consensus workflow yields a compact five-gene candidate set from 490 input variables. Its principal strength is the combination of ten methods that emphasize different forms of predictive relevance. The recurrence-based tie-breaker rewards features that repeatedly appear near the top, while the normalized consensus score summarizes the magnitude of importance on a common range. This is more transparent than choosing a panel from a single importance plot.

The feature-importance evidence should nevertheless be interpreted carefully. Min–max normalization makes scales comparable only in a limited numerical sense; it does not calibrate the reliability of each method. An ANOVA statistic, a mean absolute SHAP value, a permutation-induced macro-F1 decrease, and a sparse regression coefficient do not share the same statistical interpretation. Equal-weight averaging gives every method equal influence regardless of uncertainty. The consensus panel is thus a pragmatic ranking device, not a formal posterior probability or causal score.

The classification experiments illustrate the performance-versus-parsimony trade-off. The full feature set retains information that is lost when reducing the model to five genes. The compact panel is potentially easier to interpret and could motivate targeted assays, but the current results do not show that it can replace the full input matrix without a material loss in macro-F1. Furthermore, the seven-class target is imbalanced, and NC has only six cases. The high overall accuracy reported by the XGBoost experiment does not imply equally strong recognition across all classes.

The survival analysis is a separate exploratory question. The increase in apparent C-index after adding five genes is approximately 0.0021, which is small. Since both values are computed in-sample, the observed difference may reflect overfitting and cannot establish incremental clinical utility. The non-significant gene coefficients further argue against making independent prognostic claims from this experiment. More rigorous survival analysis should use out-of-sample risk predictions, confidence intervals, model diagnostics, and external validation.

The most important technical correction is to eliminate test-set leakage from feature selection. If the held-out test set influences SHAP, permutation scores, or consensus ranking, the reported test score is no longer an unbiased assessment of a fixed pipeline. All feature-selection steps must be nested inside training folds. This correction may change the selected genes and the measured performance, and it should be completed before publication.

8 Conclusion

This analysis ranks 490 candidate gene-related features using ten complementary selection procedures and identifies *EGFR*, *CDK1*, *GATA3*, *ERBB2*, and *ERBB3* as a compact candidate panel. The supplied notebook reports an XGBoost full-feature accuracy of 0.8346 and macro-F1 of 0.6780. In the separate model comparison, RBF-SVM achieves the highest macro-F1 on the full feature set (0.6756), while Gaussian Naive Bayes is the strongest five-gene model by the reported summary (macro-F1 0.5505). The panel reduces dimensionality by 98.98%, but this comes with a reduction in predictive performance. The apparent Cox C-index changes only from 0.6834 to 0.6855 after adding the panel, and the analysis does not establish statistically significant independent survival effects. The findings are therefore exploratory. Nested feature selection, class-imbalance-aware evaluation, out-of-sample survival assessment, and external validation are necessary before claiming a validated subtype signature or clinical value.

References

[1] C. Curtis *et al.*, "The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups," *Nature*, vol. 486, pp. 346–352, 2012.

[2] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[4] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[5] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society: Series B*, vol. 58, no. 1, pp. 267–288, 1996.

[6] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society: Series B*, vol. 67, no. 2, pp. 301–320, 2005.

[7] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. Wiley, 1991.

[8] E. L. Kaplan and P. Meier, "Nonparametric estimation from incomplete observations," *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 457–481, 1958.

[9] D. R. Cox, "Regression models and life-tables," *Journal of the Royal Statistical Society: Series B*, vol. 34, no. 2, pp. 187–220, 1972.

[10] F. E. Harrell Jr. *et al.*, "Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors," *Statistics in Medicine*, vol. 15, pp. 361–387, 1996.

[11] C. Molnar, *Interpretable Machine Learning*, 2nd ed., 2022.